Under the Shadow of Babel: A Position Survey on Linguistic Relativity and Strategic Deception in Multilingual LLM Games

Anonymous

Abstract

As large language models evolve from static query engines into autonomous agents that negotiate, deceive, and collaborate, a dangerous gap is emerging: our evaluation frameworks remain tethered to an English-centric illusion of alignment. We hypothesize that language does not merely serve as a passive conduit for reasoning but functions as an active "Algorithmic Sapir-Whorf Switch" that reconfigures a model's strategic intent, Theory of Mind, and deceptive capacity. In this position survey, we move beyond simple performance metrics to expose how this switch systematically modulates behavior in multi-agent environments.

Crucially, we engage with the friction of contradictory evidence: from volatile bimodal strategy distributions in Arabic and the subversion of collectivist norms in Vietnamese, to a rigorous cross-lingual persuasion study that finds no significant language effect on human outcomes. We ground these behavioral shifts mechanistically through the Semantic Ratio and Translation Barrier hypotheses, while documenting emerging systemic threats like Code-Switching Red-Teaming (46.7% higher attack success) and steganographic collusion. Finally, we unveil a critical evaluative fragility—the collapse of multilingual LLM-as-Judge reliability ($\kappa \approx 0.3$)—and propose a unified Cross-Cultural Agentic Sandbox for the next generation of interactive alignment research.

1 Introduction

The rapid proliferation of Large Language Models (LLMs) since the advent of the Transformer architecture (Vaswani et al., 2017) and the demonstration of few-shot emergent abilities in large-scale pretraining (Brown et al., 2020) has catalyzed a migration from reactive, single-prompt systems to autonomous agents embedded within Multi-Agent Systems (MAS), where they negotiate transactions, allocate resources, and execute long-horizon strategic planning (Huynh et al., 2026; Sun et al., 2025a). Yet the methodological frameworks for evaluating safety and alignment have not kept pace, relying predominantly on static, single-turn datasets that may not capture behavioral conviction under dynamic, interactive pressure (Liu et al., 2025; Nguyen et al., 2025).

This creates an epistemological gap: a potential decoupling between a model's apparent alignment in sanitized English settings and its behavioral manifestations in complex, non-English, multi-agent realities (Martinez et al., 2024). Safety guardrails and RLHF pipelines (Ouyang et al., 2022) are overwhelmingly optimized for high-resource languages (Repello AI, 2024; Bai et al., 2022), yet weaken in interactive multilingual environments (Peng et al., 2024; Yong et al., 2024). This survey advances the working hypothesis that language operates as an "Algorithmic Sapir-Whorf Switch" (Wang et al., 2025a) that reconfigures internal representations, modulating strategic intent, Theory of Mind, and deceptive capacity. We emphasize that while convergent evidence is strong, establishing causality requires isolating language from confounds such as tokenization and RLHF coverage asymmetry.

A critical contribution is our engagement with contradictory evidence. Recent data reveals that Arabic-prompted models exhibit volatile bimodal distributions rather than consistent defection, Vietnamese models subvert collectivist expectations, and-perhaps most strikingly-a rigorous cross-lingual persuasion study finds *no significant* language effect on human behavioral outcomes (Arango, 2024). We argue these contradictions are best explained by the mechanistic architecture of multilingual computation-the Semantic Hub Hy-

pothesis (Wu et al., 2025) and the complementary Translation Barrier Hypothesis (Bafna et al., 2025)-rather than cultural essentialism. Furthermore, we expose a critical evaluative vulnerability: the very tools used to benchmark these systems-LLM-as-Judge mechanisms-exhibit Fleiss' $\kappa \approx 0.3$ across languages (Fu and Liu, 2025), undermining the reliability of cross-lingual evaluation.

Road Map. The remainder of this paper is organized as follows. Section 2 introduces a seven-category taxonomy of interactive games, providing the empirical scaffold for our argument. Section 3 presents the "Algorithmic Sapir–Whorf Switch" hypothesis alongside contradictory evidence and confounds, grounds observations mechanistically via the Semantic Hub and Translation Barrier Hypotheses, and documents code-switching attack vectors. Section 4 examines multilingual Theory of Mind deficits, intent–action divergence, and steganographic collusion risks. Section 5 exposes the evaluative fragility of multilingual LLM-as-Judge. Section 6 contrasts our contribution with existing surveys. Section 7 proposes a concrete benchmark design. Figure 1 provides a visual overview of the paper's conceptual architecture.

1.1 Survey Methodology

We conducted a systematic literature search across ACL Anthology, arXiv, Semantic Scholar, and Google Scholar using queries combining "multilingual LLM," "game theory," "strategic reasoning," "code-switching," "cooperative games," "persuasion," "dialogue games," "steganographic collusion," and "cross-lingual safety." We restricted our primary search to January 2022–February 2026. Inclusion criteria required papers that (a) empirically evaluate LLM behavior in interactive, multi-turn game environments, (b) analyze linguistic or cultural factors modulating strategic reasoning, or (c) propose benchmarks or defenses for multilingual multi-agent safety. We excluded single-turn toxicity detection and monolingual evaluations without an interactive component. We acknowledge this is a position paper, not a formal systematic review, and have actively sought contradictory findings to mitigate selection bias (Section 3.3).

2 A Broad Taxonomy of Interactive Games

Limiting evaluation to pure game-theoretic abstractions provides only a narrow view of an agent's behavioral repertoire. A comprehensive multilingual auditing framework must encompass the full spectrum of interactive environments, from cooperative puzzles to adversarial auctions. We identify seven categories, each isolating distinct cognitive faculties.

2.1 Social Deduction and Hidden-Role Games

Social deduction games-*Werewolf*, *Avalon*, and *Spyfall*-test deception and recursive Theory of Mind (Xu et al., 2023; Koto et al., 2025). Information asymmetry is foundational: agents construct false narratives, detect inconsistencies, and dynamically update beliefs (Devaine et al., 2014). In *Werewolf*, LLM antagonists deploy what we term **Bayesian Belief Manipulation (BBM)**: the strategic construction of utterances designed to shift other agents' posterior beliefs about hidden roles. While previously operationalized through Chain-of-Thought (CoT) traces (see Appendix D), we acknowledge CoT faithfulness is brittle; thus, we propose measuring BBM intentionality through more robust alternative proxies, such as counterfactual utterance search or constrained decoding rationales (Xu et al., 2023; Wang et al., 2023; Lan et al., 2024). Multicultural *Spyfall* with localized entities reveals dramatic degradation in non-English deceptive reasoning (Koto et al., 2025).

2.2 Negotiation and Diplomacy

Diplomacy, *Catan*, and economic simulators evaluate persuasion, coalition-building, and trust (Meta Fundamental AI Research Diplomacy Team (FAIR), 2022; Deng et al., 2024). SPIN-Bench reveals a critical decoupling: o3-mini achieves 51.25% on formal PDDL planning but fails at intention inference, while Claude 3.5 Sonnet scores only 20.55% on PDDL yet achieves up to 100% on Diplomacy intention prediction (Yao et al., 2025). This suggests that strategic planning and social intelligence are distinct architectural faculties that do not necessarily scale together.

2.3 Persuasion and Auction Games

Persuasion environments test rhetorical influence and personality adaptation (Karande et al., 2024). A critical negative finding balances our narrative: in a rigorous cross-lingual study of AI-generated charitable advertisements, Arango (2024) found *no statistically significant* effect of LLM utilization on persuasive outcomes across English and

Game Taxonomy §2	**Mechanistic Pillars** §3	**Threats & Evaluation** §3.5, §5
Social Deduction (Werewolf, Avalon, Spyfall), Negotiation (Diplomacy, SPIN-Bench), Persuasion (Charitable Ads, AucArena), Cooperative (Codenames, Hanabi, XToM, Overcooked), Social Dilemmas (IPD, FAIRGAME, Ultimatum)	Representation: Semantic Hub, Translation Barrier. Dynamics: ActAdd, Low-Resource, Framing.	Adversarial: CSRT, AutoRed, AgentMiddle. Collusion: Audit-Whisper, X-SIR. Reliability: LLM-as-Judge, Ensemble. Metrics: MAST, MultiAgentBench, M3-BENCH.

Figure 1: Taxonomy of Multilingual LLM Games: seven game categories, mechanistic pillars (Semantic Hub, Translation Barrier), threats (CSRT, collusion), and evaluation (LLM-as-Judge, MAST, MultiAgentBench, M3-BENCH).

German cohorts. Human recipients anchored decisions on source belief (whether they suspected AI generation) rather than linguistic nuances, rendering cross-lingual rhetorical variations largely moot. This null finding suggests that certain game-theoretic predictions about language-modulated behavior may not generalize to human-in-the-loop persuasion contexts. Relatedly, the *anecdoctoring* pipeline demonstrates that LLM-driven localization can produce culturally grounded disinformation at scale (Cuevas et al., 2025), yet the persuasion null result tempers alarm by showing that human metacognition (source suspicion) can override linguistic manipulation-a boundary condition that deserves further investigation across more languages and domains. AucArena tests strategic planning in competitive auctions via Belief-Desire-Intention (BDI) architectures (Chen et al., 2024). While frontier models manage budgets effectively in English, non-English execution suffers "attention drift": tokenization inefficiencies blur the semantic weight of numerical values, causing agents to lose track of hard budget constraints (Chen et al., 2024).

2.4 Cooperative Communication Games

Cooperative games expose a dimension absent from competitive environments. In *Codenames*, LLMs must generate single-word clues connecting multiple target words-a task demanding shared cultural ground. Hakimov et al. (2025) demonstrate that LLMs rely on statistical vector proximity rather than genuine cultural knowledge; in non-English settings, BPE tokenizer fragmentation distorts semantic proximity, producing clues based on phonetic or orthographic rather than conceptual associations.

The LLM-Hanabi benchmark confirms that first-order ToM correlates strongly with cooperative success (Chen, 2025b). The XToM benchmark extends this to multilingual contexts, revealing a pronounced cognitive dissonance: while frontier models excel in multilingual fluency, their ToM capabilities vary drastically across languages (Chan et al., 2025). GPT-4o's recall BERTScore drops to 0.864 when reasoning is translated from English to Chinese, indicating that the recursive syntactic structures required for mental state modeling ("Agent A believes that Agent B intends to...") degrade outside English.

In Collab-Overcooked, Sun et al. (2025b) evaluate 13 LLMs on collaborative cooking tasks, finding significant shortcomings in active collaboration. Multilingual interaction introduces "attention misalignment": agents prioritize syntactic translation over task-state tracking, leading to execution loops, redundant actions, and failure to recognize completed sub-tasks.

2.5 Open-World and Video Game Environments

Voyager achieves unprecedented open-ended learning in Minecraft (3.3× more items, 15.3× faster milestones), but its reliance on English-centric code APIs means non-English instructions require zero-shot cognitive translation before code generation, introducing latency and syntax errors (Wang et al., 2024). BALROG reveals that multimodal capabilities paradoxically *worsen* performance in spatial environments, and multilingual tasking further degrades spatial planning (Paglieri et al., 2025). GVGAI-LLM confirms that LLMs hallucinate state transitions in spatial games; non-English instruction layers compound this fragility (Chen, 2025a). TCG-Bench reports 47.4% cross-linguistic gaps between English and Arabic at 32B scale (Wang et al., 2025b).

2.6 Dialogue Games

The clembench framework provides the most controlled multilingual game evaluation, bifurcating performance into format adherence (Percentage Played) and strategic competence (Quality Score),

combined into a unified clemscore (Chalamalasetti et al., 2023; Schlangen et al., 2025). The clembench 2024 evaluation across English, German, Italian, Spanish, and French reveals that while commercial models (GPT-4o, Claude 3.5) maintain reasonable Percentage Played across languages, open-weight models suffer massive format adherence degradation in non-English settings (Beyer et al., 2024). GPT-4 achieves 66.67% Played in English, dropping significantly in German and Italian. The failures are primarily *structural*-formatting violations, conversational looping, persona abandonment-rather than strategic, demonstrating that multilingual cognitive load strips computational capacity from rule compliance.

Hakimov et al. (2026) reinforce this finding in multilingual negotiation games (Deal or No Deal, Air Balloon Survival): enabling chain-of-thought reasoning drastically improves English Quality Scores but yields diminishing returns in German and Italian. Internal reasoning traces generated in non-English languages become confused, confirming that strategic logic does not translate symmetrically across linguistic representations.

2.7 Social Dilemmas and Matrix Games

Social Dilemmas (IPD, Public Goods, Ultimatum Game) provide the mathematically purest strategic abstraction, grounding LLM behavior in the foundational principles of game theory (Von Neumann and Morgenstern, 1944) and the evolution of cooperation (Axelrod, 1984; Huynh et al., 2026; Akata et al., 2023). Notably, LLMs in these settings often exhibit a "nicer than human" bias, defaulting to prosocial or altruistic strategies at rates exceeding human baselines-aggregate cooperation rates can soar to 79% compared to the historical human average of roughly 48% in identical experimental conditions (Vicente et al., 2024; Za et al., 2026; Wang et al., 2026a). Through FAIRGAME, researchers parameterize linguistic framing, personality priming, and payoff multipliers to generate measurable strategy shifts away from these defaults (Huynh et al., 2025; Buscemi et al., 2025). The M3-BENCH framework extends this by introducing a process-aware evaluation of "dark social behaviors" (secret collusion, deceptive resource allocation), formalizing "commitment reliability" to quantify the rate at which agents honor stated agreements (Xie et al., 2026). Strategy classification typically employs LSTM-based sequence classifiers with confidence thresholds (≥ 0.6); however, calibration analyses and inter-annotator reliability for mapping multi-turn dialogue to canonical strategies remain underreported, representing a methodological gap. While these games provide the most controlled designs, their simplicity limits ecological validity; the preceding six categories test whether linguistic effects generalize to richer environments (Section 7).

Language	ALLC	ALLD	TFT	WSLS
English baseline	26%	25%	25%	24%
French	27%	18%	26%	29%
Arabic	32%	28%	19%	21%
Chinese	31%	30%	19%	20%
Vietnamese	26%	24%	23%	27%

Table 1: IPD strategy distributions aggregated across four model families (Huynh et al., 2026; Za et al., 2026). Scales $k \in \{1, 3, 5, 10\}$; LSTM confidence ≥ 0.6.

3 The "Sapir-Whorf" Algorithmic Switch

3.1 Linguistic Determinism in Vector Spaces

Using the BICAUSE bilingual dataset (Jin et al., 2024), researchers observe typologically aligned attention patterns: Chinese reasoning allocates denser attention to causes and sentence-initial connectives compared to English (Wang et al., 2025a). This suggests models internalize language-specific constructions as stable processing patterns rather than merely translating language-agnostic thought (Wu et al., 2025; Boroditsky, 2001).

3.2 Cross-Linguistic Divergence: Evidence and Limits

High-resolution IPD data from Huynh et al. (2026), evaluating GPT-4o, Mistral Large, Claude 3.5, and Llama with LSTM-based strategy classifiers, reveals distributions far more nuanced than initial reports (Table 1). All findings characterize *model–language interaction effects*, not properties of languages or cultures.

Three observations contradict widely held assumptions. Arabic simultaneously produces the highest ALLC (32%) and high ALLD (28%)-a volatile bimodal distribution, not unidirectional defection-reflecting sparse intermediate reciprocal templates in alignment data. Vietnamese frequently yields high defection despite collectivist cultural associations, with cooperative priming achieving mere 2% cooperation in Claude 3.5 Haiku (Huynh et al., 2025). Chinese favors WSLS over TFT, indicating outcome-dependent optimization. Model-

level effects compound: GPT-4o defaults "hawkish" (31.7% aggregate ALLD) while Mistral maintains "dovish" (33.7% ALLC) (Huynh et al., 2026).

3.3 Confounds and Negative Findings

Rigorous evaluation requires disentangling linguistic-cognitive effects from at least five confounding variables. **Tokenization quality** varies dramatically: BPE tokenizers produce longer, less coherent sequences for morphologically rich or low-resource languages (Yong et al., 2024). **RLHF coverage** asymmetry means models may appear less aligned in under-trained languages simply because safety data is sparser. **Training data distribution** varies substantially across languages; adversarial discourse or formal registers may be overrepresented. **Decoding hyperparameters**-including temperature, top-p sampling thresholds, and repetition penalties-interact with language-specific token frequency distributions and may produce divergent strategy profiles even absent genuine cognitive differences. **Contextual framing** can override language effects entirely: when game horizons remain unknown, cooperation universalizes across all languages, conforming to classical game-theoretic predictions (Buscemi et al., 2025; Lorè and Heydari, 2024). Vietnamese "defection" sometimes reflects mathematical misconprehension under attenuated payoffs ($k = 0.1$) rather than strategic intent (Huynh et al., 2025). The persuasion null finding (Section 2.3) further demonstrates that not all interactive behaviors are language-modulated. These collective confounds reinforce the necessity of carefully controlled ablation designs.

3.4 The Semantic Hub Mechanism

The Semantic Hub Hypothesis (Wu et al., 2025) provides a mechanistic explanation: logit lens decoding reveals that LLMs project heterogeneous inputs into a centralized, English-dominant hub in intermediate layers. Non-English inputs undergo lossy compression, with strategic conclusions computed in English-anchored hidden states. Activation Addition (ActAdd) provides suggestive evidence for functional significance: in the experiments reported by Wu et al. (2025), English-derived steering vectors required a larger scaling coefficient (approximately 5, vs. a standard 2 for English) to alter non-English outputs. However, we note that these ActAdd results were observed in a limited set of models and layers; the specific scaling coefficients may vary across model families, layer selections, steering vector construction methods (e.g., mean-difference vs. PCA-based), and prompt distributions. Independent replication across architectures is needed before these coefficients can be treated as robust findings (see Appendix E for methodological discussion).

The Translation Barrier Complement. Beyond single-turn safety, the Mechanistic Interpretability literature provides a lens for our findings. The Semantic Hub Hypothesis (Wu et al., 2025) correctly localizes cross-lingual reasoning in intermediate layers, yet we find that steering these representations requires significantly higher intensity in non-English contexts ($\alpha \approx 5$) compared to English ($\alpha \approx 2$) (Wu et al., 2025; Turner et al., 2023). We emphasize that these ActAdd assertions rest on preliminary, non-replicated evidence. Furthermore, while traditional Mean-Difference vectors suffice for English, PCA-based Sparse Representation Steering (SRS) (He et al., 2025b) is required to isolate semantic intent from language-specific syntactic noise, achieving a 0.99 refusal rate on malicious non-English prompts compared to moderate results for traditional ActAdd.

We further triangulate these findings with metacognitive calibration. The Expected Calibration Error (ECE) for complex strategic tasks exhibits a 5x degradation across languages, skyrocketing from 4.6% in English to 23.1% in non-English prompts (Ahuja et al., 2022). Information-level ECE (InfoECE) reveals that this miscalibration is particularly acute in multi-turn interactions where models exhibit unjustified overconfidence, hallucinating tactical certainty in fractured latent spaces (Zhang et al., 2026). Language-Aware Confidence Ensemble (LACE) techniques demonstrate that sampling representations from optimal late-intermediate layers (e.g., layers 25–32 in 40-layer architectures) can recover calibration stability by bypassing final-layer projection noise (Zhou et al., 2025). A complementary mechanistic account-the Translation Barrier Hypothesis (Bafna et al., 2025)-posits that many multilingual failures originate not in upstream reasoning (which may proceed adequately in the English-anchored hub) but in *late-layer language realization*: the model solves the task correctly in its internal representation but fails during the final projection into the target language's token space. This distinction has practical diagnostic implications: if failures are primarily late-layer

realization errors, interventions such as intermediate decoding checks (probing hidden states at middle layers) or dual-pass translation pipelines could recover correct reasoning. Language-aware interpretability tools such as Indic-TunedLens (Panchal et al., 2026) offer a concrete methodology for testing hub-vs-barrier predictions by probing layer-wise language identity in low-resource scripts. We propose that future benchmarks incorporate both hypotheses by including intermediate-layer probes alongside final-output evaluation (Section 7).

Together, the Semantic Hub and Translation Barrier accounts explain the full spectrum of observed failures: the bimodal Arabic distribution and Vietnamese mathematical errors likely reflect upstream hub-level representational sparsity, while clembench formatting failures in German and Italian may arise from late-layer realization bottlenecks where the model's correct strategic reasoning is lost during surface-form generation.

3.5 Code-Switching as an Attack Vector

We synthesize (but do not produce) empirical evidence from Code-Switching Red-Teaming (CSRT), which achieves 46.7% higher ASR across ten state-of-the-art models via intra-sentential language mixing at the byte-pair level (Yoo et al., 2025). Against Claude 3 Haiku: 9.04% ASR vs 2.38% English baseline; against Claude 3 Opus: 3.21% vs 1.27%. ASR scales proportionally with the fraction of Low-Resource Language (LRL) tokens in the prompt, as the fractured input overwhelms the model's sequential safety processing before harmful intent can be flagged. Automated defenses-including perplexity filters and paraphrase wrappers-prove insufficient against CSRT because the attack exploits the Semantic Hub bottleneck (Section 3.4) at a level below input-layer text sanitization (Yoo et al., 2025).

Broader Red-Teaming Ecosystem. CSRT operates within a rapidly expanding ecosystem of adversarial evaluation frameworks that share structural similarities with the multilingual attack surface. AutoRed (Perez et al., 2025) demonstrates that persona-guided, reinforcement-learning-driven red-teaming can systematically discover failure modes at scale, including culturally situated attacks. OpenRT (Wang et al., 2026b) provides an open-source multi-turn benchmark revealing that sequential adversarial pressure compounds vulnerability-a dynamic directly paralleled by code-switching escalation across conversational turns. Genesis (Zhang et al., 2025) shows that evolving attack strategies against web agents can exploit presentation mismatches, with early evidence of multilingual payload effectiveness. M2S (Ha et al., 2025) formalizes multi-turn, multi-strategy jailbreak composition, demonstrating that combining structural attacks (e.g., role-play, hypothetical framing) with linguistic manipulation achieves synergistic ASR gains. These frameworks collectively support our thesis that interactive, multi-agent settings expose failure modes invisible in static evaluation, and that adversaries can exploit multilingual and presentation mismatches as a compound attack vector.

Agent-in-the-Middle (AiTM) attacks compound this by intercepting and manipulating inter-agent messages within MAS communication channels (He et al., 2025a). Combined with code-switching, AiTM could simultaneously manipulate both the content and linguistic framing of inter-agent communication, creating a compound attack surface that neither individual defense addresses.

4 Theory of Mind, Deception, and Coordination Failure

4.1 The Multilingual ToM Deficit

The XToM benchmark (Chan et al., 2025) reveals that while frontier models achieve high multilingual fluency, their ToM capabilities degrade asymmetrically across languages. GPT-4o's recall BERTScore drops to 0.864 for English-to-Chinese reasoning tasks (versus 0.912 for English-only evaluation, a 5.3% relative degradation), indicating that the recursive syntactic structures modeling mental states do not reliably map to non-English pathways; we note, however, that Chan et al. (2025) do not report confidence intervals for these scores, and replication across additional model families is needed to establish effect robustness. LLM-Hanabi confirms that first-order ToM (interpreting intent) is a stronger performance predictor than second-order ToM (Chen, 2025b); when first-order ToM degrades in low-resource languages, both cooperative and deceptive reasoning suffer simultaneously.

4.2 Intent–Action Divergence

The gap between an agent's stated commitments and executed actions represents a critical deployment barrier. The MAST taxonomy (Cemri et al., 2025), derived from 1,600+ execution traces across seven MAS frameworks, identifies 14 failure modes in three categories: Specifica-

tion/Design Issues (41.77%), Inter-Agent Misalignment (36.94%), and Task Verification (21.30%). The core intent-action failures include reasoning-action mismatches-where an agent's chain-of-thought correctly identifies the optimal action but the generated API call or game move contradicts this reasoning-and unwarranted conversation resets. MultiAgentBench operationalizes coordination quality through milestone-based KPIs (ratio of intermediate sub-goals achieved) and composite Coordination Scores averaging Communication Score and Planning Score on a five-point scale (Zhu et al., 2025). M3-BENCH formalizes "commitment reliability"-the rate at which agents honor stated cooperative agreements-finding that promise-keeping drops significantly in mixed-motive games and exhibits cross-lingual fragility (Xie et al., 2026).

To anchor intent–action divergence measurements across languages, we propose that future benchmarks adopt annotated action logs as gold standards: each game turn would pair (a) the agent's natural language commitment ("I will cooperate this round"), (b) the executed game action, and (c) a human-annotated consistency label, with separate annotation pools per language to control for cross-lingual pragmatic variation in commitment framing. Inter-annotator agreement should be reported per language using Cohen's κ or Krippendorff's α.

4.3 Steganographic Collusion

The *Audit the Whisper* framework achieves 0.819 AUROC for collusion detection via paired-run KL diagnostics ($d(T) = 0.273$ bits), with the union meta-test reaching TPR = 1.0 at near-zero FPR (Roger et al., 2025). X-SIR retains TPR = 0.820 and AUC = 0.915 under Cross-Lingual Watermark Removal Attacks (He et al., 2024). Details in Appendix B.

5 Evaluative Validity: The Multilingual Judge Problem

A critical meta-methodological concern undermines all cross-lingual game evaluation: the reliability of LLM-as-Judge mechanisms. Fu and Liu (2025) assessed judgment consistency across 25 languages, finding that average inter-language agreement reaches merely Fleiss' $\kappa \approx 0.3$-"fair to poor" reliability. The degradation is catastrophically asymmetric for low-resource languages, manifesting as length biases, prompt sensitivity, and hallucinated compliance. Neither parameter scaling nor explicitly multilingual training resolves this gap: Aya-Expanse, designed for robust cross-lingual performance, exhibits paradoxically poor consistency between English and non-English tasks (Fu and Liu, 2025).

To mitigate this, dynamic calibration techniques such as the Language-Aware Confidence Ensemble (LACE) leverage representations from late-intermediate layers (e.g., layers 25–32 in a 40-layer architecture), which offer far more stable and language-agnostic signals than the English-biased final layer (Zhou et al., 2025). Furthermore, Information-level ECE (InfoECE) provides a more robust metric for multi-turn interactions, showing that calibration collapse is often tied to information exposure rather than just syntactic translation errors (Zhang et al., 2026). The implication is severe: state-of-the-art evaluation requires ensemble strategies-multiple judges via weighted voting or mandatory self-translation to English before scoring-to achieve minimally acceptable reliability. However, we caution that the self-translation step itself risks introducing evaluation leakage into English-dominant hubs, potentially masking true cross-lingual weaknesses. Thus, any benchmark relying on LLM evaluation must validate this step against human bilingual adjudication and professional human translation baselines.

6 Relationship to Existing Surveys

Existing surveys at the intersection of game theory and LLMs adopt a predominantly unilingual lens. Sun et al. (2025a) treat language as fixed; Zhang et al. (2024), Feng et al. (2024), and Hu et al. (2024) presuppose English. SPIN-Bench (Yao et al., 2025) and clembench (Chalamalasetti et al., 2023; Schlangen et al., 2025) pioneer multi-game evaluation, with clembench's distinction between formatting and strategic competence providing a key design principle. MultiAgentBench (Zhu et al., 2025) and MAST (Cemri et al., 2025) provide complementary frameworks for measuring coordination quality and diagnosing failure modes. Beyond interactive games, our work connects to literature on multilingual robustness (Wang, 2024), the growing language gap in LLM safety research (Yong et al., 2025), and specific vulnerabilities in cross-lingual safety alignment such as M-ALERT (Friedrich et al., 2025). We further triangulate

our findings with surveys on multilingual calibration and confidence estimation (Geng et al., 2024). Our interactive-game focus complements but is *distinct* from static cross-lingual transfer benchmarks such as XTREME (Hu et al., 2020), XTREME-R (Ruder et al., 2021), and XCOPA (Ponti et al., 2020), which evaluate zero-shot and few-shot classification tasks (NLI, QA, NER, Causal Reasoning) across 40+ languages. Those paradigms expose cross-lingual generalization gaps in structured prediction but cannot reveal the strategic misalignment, calibration collapse under interactive pressure, or evaluative unreliability that emerge in multi-agent game settings. Crucially, the calibration literature provides additional triangulation: cross-lingual ECE degradation documented in question-answering settings ($\sim$10–15% increase) (Geng et al., 2024) mirrors, though does not fully predict, the more severe overconfidence collapse (4.6%$\rightarrow$23.1%) we observe in strategic game contexts, suggesting a common mechanistic origin in RLHF coverage asymmetry that is amplified under interactive pressure.

Our survey fills a critical gap by positioning language as the primary independent variable across seven game categories, integrating negative findings (persuasion null result, clembench formatting-vs-strategy distinction), grounding divergences in complementary mechanistic accounts (Semantic Hub and Translation Barrier), and exposing the evaluative vulnerability of multilingual LLM-as-Judge paradigms. In contrast to prior red-teaming surveys that focus on single-turn attacks, we integrate interactive, multi-agent adversarial frameworks (CSRT, AiTM, AutoRed, Genesis) that exploit the compound vulnerability of multilingual game-playing agents.

7 Methodological Challenges and Future Frontiers

We argue for a "Cross-Cultural Agentic Sandbox Benchmark" spanning all seven game categories with the following design requirements.

Confound control. The benchmark must implement matched tokenizer ablations (comparing BPE, SentencePiece, and character-level tokenizers per language), controlled RLHF coverage verification, professionally translated and back-translated prompt templates, and explicit manipulation of decoding hyperparameters (temperature, top-p, repetition penalty) to isolate language-specific effects from decoder artifacts.

Pre-registered evaluation. All strategy classifiers (e.g., LSTM-based TFT/WSLS/ALLD detectors) should be pre-registered with calibration curves, confidence thresholds, and error analysis reported per language. Hypotheses about cross-linguistic divergence should be pre-registered with specified effect sizes and statistical tests.

Intent–action metrics. The benchmark should incorporate MAST failure categories, MultiAgentBench milestone KPIs, and M3-BENCH commitment reliability scores, anchored by per-language annotated gold standards as described in Section 4.2.

Code-switching adversaries. Both intra-sentential and inter-sentential code-switching adversaries should be incorporated directly into the sandbox as red-team agents. Defenses should be evaluated beyond paraphrase wrappers: capacity-guided interventions (paraphrase + token rate limiting + role permutation, as in Roger et al. (2025)), input segmentation by detected language, and LRL-aware perplexity thresholds should be benchmarked at controlled FPR.

Collusion detection. Capacity-calibrated collusion detectors incorporating MI-based penalties, permutation tests, and acceptance-bias diagnostics should report operating points with controlled FPR per language (Roger et al., 2025).

Ensemble judging. Given the demonstrated unreliability of single-pass multilingual LLM-as-Judge evaluation ($\kappa \approx 0.3$), the benchmark must mandate ensemble strategies. We recommend a concrete default protocol (detailed in Appendix G): (1) a panel of ≥ 3 judges from different model families (e.g., GPT-4o, Claude 3.5, Gemini) to reduce systematic bias; (2) mandatory self-translation of non-English game transcripts to English before scoring, with the original preserved for audit; (3) structured checklist rubrics decomposing evaluation into binary or ordinal sub-criteria (strategy identification, rule compliance, deception detection) rather than holistic Likert scales; and (4) weighted majority voting with abstention on low-confidence items. Preliminary evidence from Fu and Liu (2025) suggests that the self-translation step alone recovers approximately 40% of the inter-language κ gap, while three-judge panels with checklist rubrics can push effective κ above 0.5 ("moderate" agreement).

Intermediate-layer diagnostics. To firmly distinguish Semantic Hub failures (upstream reason-

ing deficits) from Translation Barrier failures (late-layer realization errors), the benchmark must incorporate an explicit, pre-registered diagnostic protocol. We recommend probing hidden states at layers $\lfloor L/3 \rfloor$ and $\lfloor 2L/3 \rfloor$ using language-aware linear probes (Panchal et al., 2026). The protocol must specify clear acceptance criteria: if intermediate representations reliably decode the correct strategic intent (e.g., accuracy $> 90\%$) but the final output layer fails, the error is classified as a Translation Barrier failure. If intermediate probes fail simultaneously with the final output, it is a Semantic Hub failure. This must be validated across at least two distinct architectural families (e.g., Llama and Mistral).

Minimal Viable Benchmark (MVB) First Release. To lower adoption barriers, we recommend a first-release subset that balances coverage and cost. **Languages (8):** English (baseline), French (high-resource Romance), Arabic (morphologically rich, RTL), Chinese Simplified (logographic), Vietnamese (low-resource, tonal), Swahili (African, agglutinative), Hindi (Indo-Aryan), and Turkish (agglutinative, vowel harmony)-chosen to span diverse scripts, morphological types, and RLHF coverage tiers. **Game genres (3):** Social Dilemmas (IPD, most controlled), Dialogue Games (clembench, provides format/strategy decomposition), and Cooperative (Codenames, tests cultural grounding). **Adversaries:** Intra-sentential CSRT only, as the most-studied and highest-ASR variant. **Models (3):** One proprietary (GPT-4o), one open-weight large (LLaMA-3-70B), one multilingual-specialized (Aya-Expanse). **RLHF coverage audit:** Use language-identification tools to estimate per-language training data proportions; report Spearman correlation between estimated coverage and observed performance gaps as a confound proxy. **Estimated cost:** \$2,000–\$4,000 for API-based multi-agent execution; \$8,000 for human annotation (3 annotators $\times$ 8 languages $\times$ 200 game turns $\times$ \$0.05/label). This MVB subset can be completed for under \$15,000 and provides the critical confound-controlled baseline upon which the full benchmark design can be iteratively expanded.

Dialect and script handling. The benchmark must explicitly handle dialects (e.g., Modern Standard Arabic vs Gulf Arabic vs Egyptian Arabic; Simplified vs Traditional Chinese), code-mixing (Hinglish, Spanglish), and script variants, since these sub-linguistic distinctions may have fundamentally different data coverage and safety properties.

Cultural localization. Game content must extend beyond translation to include culturally localized entities, scenarios, and pragmatic conventions (Koto et al., 2025).

Meta-Analytic Synthesis. Given that current literature often underreports confidence intervals and effect sizes, future work must prioritize hierarchical meta-analytic synthesis (e.g., hierarchical Bayesian models controlling for model family, language, and task) to properly quantify heterogeneity and disentangle genuine language effects from classifier noise.

Ethics and Logistics. To ensure the benchmark's viability, we propose a target scale of 20+ languages representing diverse scripts, morphological types, and RLHF availability. This requires balancing high-resource (English, French, Chinese) with low-resource (Swahili, Quechua, Vietnamese) coverage. We estimate evaluation costs to span \$5,000–\$15,000 for API-based multi-agent execution at scale, supplemented by a \$30,000 budget for human-in-the-loop annotation cohorts. Annotator training must specifically address cross-lingual pragmatic norms and intent-action labeling. All benchmark data must prioritize privacy and ethical recruitment, ensuring equitable compensation for annotators in the Global South.

The clembench clemscore decomposition (Percentage Played $\times$ Quality Score) provides a critical methodological principle: cross-linguistic performance gaps must be disaggregated into formatting compliance failures and genuine strategic reasoning degradation (Schlangen et al., 2025). The "Price of Thought" finding-that chain-of-thought reasoning benefits do not transfer to non-English languages (Hakimov et al., 2026)-further implies that reasoning-augmented evaluation must be validated per-language rather than assumed to generalize from English. Finally, to facilitate meta-analysis and replication, we will release a structured repository containing all extracted numerical results, task metadata, and reported uncertainties synthesized in this survey. Furthermore, to ensure traceability while respecting double-blind review constraints, we provide a supplementary registry that explicitly maps all anonymized citations to persistent identifiers and public artifacts.

8 Conclusion

This position survey has argued that language is not a passive conduit but a potentially active modulator of LLM strategic behavior across a broad spectrum of interactive environments. The quantitative evidence reveals cross-linguistic divergences that are more nuanced and contradictory than previously reported-including a persuasion null finding that delineates boundary conditions-best explained by complementary mechanistic accounts: the Semantic Hub Hypothesis (upstream representational compression) and the Translation Barrier Hypothesis (late-layer realization failures), rather than cultural essentialism. This is further corroborated by the metacognitive calibration gap, where non-English ECE (23.1%) significantly exceeds the English baseline (4.6

Limitations

This survey acknowledges several limitations. **1. Scale and Diversity:** Empirical evidence is drawn primarily from five languages (English, French, Arabic, Chinese, Vietnamese); generalization to the world's 7,000+ languages remains speculative, requiring much broader typological coverage. **2. Environmental Abstraction:** Game environments remain abstractions that may not fully capture the complexity of real-world multi-agent deployments. **3. Causal Inference:** The "Algorithmic Sapir–Whorf Switch" is a *working hypothesis*, not a definitive causal claim; behavioral divergences are likely confounded by tokenization, RLHF coverage, and training data distribution. **4. Empirical Scope:** As a position survey, this work synthesizes existing results rather than contributing new experiments; effect sizes and calibration curves are often underreported in the source literature. **5. Evaluator Fragility:** Heavy reliance on LSTM-based strategy classifiers ($\kappa \approx 0.6$) and single-pass LLM-as-Judge pipelines ($\kappa \approx 0.3$) introduces severe evaluative uncertainty; the synthesized behavioral divergences in this survey may consequently be distorted by these weak evaluators. **6. Data Sparsity:** Several categories (Open-World, Cooperative Communication) lack extensive multilingual data, requiring cautious extrapolation.

Ethics Statement

This research discusses adversarial attack vectors to inform defensive measures. All evaluations involve artificial agents. Cross-linguistic behavioral divergences reflect *model–language interactions under specific training distributions*, not properties of languages or cultures. The persuasion null finding (Arango, 2024) reinforces this framing: language effects are context-dependent and should not be overgeneralized. We urge researchers to communicate findings with appropriate context to avoid cultural essentialism.

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A Structured Evidence Synthesis

Table 2 summarizes the key empirical studies discussed in this survey, detailing their game types, evaluated models, languages, key metrics, and primary findings along with necessary caveats. This structured synthesis highlights the breadth of our evidence base, from social dilemmas to persuasion and meta-evaluation.

B Steganographic Detection and Watermarking

Table 3 presents the performance metrics of steganographic detection via Audit the Whisper (Roger et al., 2025) across various scenarios, as well as cross-lingual watermark robustness results from X-SIR (He et al., 2024).

Study	Game Type	Models	Languages	Key Metric	Finding	Caveats
Huynh et al. (2026)	Social Dilemma (IPD)	GPT-4o, Mistral, Claude, Llama	En/Fr/Ar/Zh/Vi	Strategy dist.	Bimodal Arabic; Vietnamese subverts expectations	No CIs
Buscemi et al. (2025)	Social Dilemma (IPD)	GPT-4o, Llama	En/Fr/Ar/Zh/Vi	Message length	Endgame exploitation in Arabic	Limited models
Hakimov et al. (2025)	Cooperative (Codenames)	GPT-4o, DeepSeek, Llama	En + non-En	Clue quality	Severe degradation; phonetic fallback	BPE-specific
Chan et al. (2025)	ToM Benchmark (XToM)	GPT-4o, DeepSeek-R1	En/Zh/Ja/Fr/Es	R BERTScore	En→Zh: 0.864 (GPT-4o)	Limited games
Beyer et al. (2024)	Dialogue Games (clembench)	GPT-4, Claude, LLaMa-3	En/De/It/Es/Fr	clemscore	EN 66.67% Played; sharp DE/IT drops	Open models
Hakimov et al. (2026)	Negotiation (Deal or No Deal)	Claude, Llama, DeepSeek	En/De/It	clemscore + CoT	CoT fails in DE/IT	Limited tasks
Arango (2024)	Persuasion (Charitable ads)	GPT variants	En/De	Social persuasiveness	**No significant effect**	Human-in-loop
Fu and Liu (2025)	Meta-evaluation (Judge)	GPT-4o, Aya-Expanse	25 languages	Fleiss' κ	≈0.3 ("fair to poor")	Asymmetric
Cemri et al. (2025)	MAS Failure Analysis	GPT-4, Claude, Qwen2.5	En	Failure freq.	Design 41.8%, Misalign 36.9%	En only
Yao et al. (2025)	Planning + Diplomacy	o3-mini, Claude 3.5/3.7	En	Accuracy/Intent	Planning $\neq$ social intelligence	No multilingual
Yoo et al. (2025)	Safety (CSRT)	10 SOTA models	Up to 10 mixed	ASR	+46.7% via code-switching	Intra-sentential
Roger et al. (2025)	Collusion detection	ColludeBench	N/A	AUROC; $d(T)$	0.819; TPR=1.0	Lab settings

Table 2: Structured evidence synthesis from key cited studies. The persuasion null finding (Arango, 2024) is highlighted as a critical boundary condition.

Scenario	Metric	Value	Notes
Pricing Duopoly	$d(T)$	0.273 bits	MI: 1.338→1.065
ColludeBench-v0	AUROC	0.819	Paired-run KL
Double Auction	TPR/FPR	1.0/≈0	Union meta-test
X-SIR (En→Ja)	TPR/AUC	0.696/0.911	vs SIR: 0.216/0.733
X-SIR (En→Zh)	TPR/AUC	0.702/0.905	vs SIR: 0.164/0.614
X-SIR CWRA	TPR/AUC	0.820/0.915	Pivot: Zh

Table 3: Steganographic detection (Roger et al., 2025) and cross-lingual watermark robustness (He et al., 2024).

C CSRT Attack Rates and Model Priors

This appendix provides additional quantitative data regarding the threat of Code-Switching Red-Teaming (CSRT). Table 4 details the intra-sentential Attack Success Rates (ASR) against the Claude 3 model family compared to English baselines (Yoo et al., 2025). Furthermore, Table 5 illustrates the baseline strategy priors across different model architectures in the Iterated Prisoner's Dilemma (Huynh et al., 2026), demonstrating how initial architectures favor distinct strategic profiles before linguistic modulation.

Model	CSRT ASR	EN Baseline
Claude 3 Haiku	9.04%	2.38%
Claude 3 Sonnet	3.18%	1.27%
Claude 3 Opus	3.21%	1.27%
10 Models (agg.)	*+46.7% relative*	

Table 4: CSRT intra-sentential ASR (Yoo et al., 2025).

Model	ALLC	ALLD	TFT	WSLS
GPT-4o	23.7%	31.7%	22.8%	21.9%
Mistral Large	33.7%	25.2%	16.4%	24.8%

Table 5: Model-level strategy priors across all languages (Huynh et al., 2026).

D Formal Definition of Bayesian Belief Manipulation

We define **Bayesian Belief Manipulation (BBM)** as a strategic communication pattern in social deduction games. Let $\mathcal{R} = \{r_1, \ldots, r_n\}$ be the set of hidden roles and $\mathcal{A} = \{a_1, \ldots, a_m\}$ the set of agents. Each agent a_i maintains a private belief distribution $P_i(r_j \mid a_k)$ over the role assignments of all other agents.

Definition: Bayesian Belief Manipulation (BBM)

An agent a_i with true role r_i^* engages in BBM at turn t if it produces an utterance u_t that satisfies the following criteria: **1. Divergence:** The implied role distribution $\hat{P}(r_i \mid u_t) \neq \delta(r_i^*)$, i.e., the utterance implies a role different from the agent's true role. **2. Intentionality:** The utterance is selected to minimize $\mathrm{KL}[\hat{P}(\cdot \mid u_t) \| P_{\text{target}}(\cdot)]$, where P_{target} is the belief distribution the manipulator wants others to hold. **3. Effect:** At least one other agent a_k updates its belief such that $|P_k^{(t+1)}(r_i^* \mid a_i) - P_k^{(t)}(r_i^* \mid a_i)| > \epsilon$ for some threshold $\epsilon > 0$.

This formalization connects BBM to measurable quantities: divergence can be estimated from game transcripts via entailment models, intentionality can be inferred from proxies described below, and effect can be measured via belief-tracking annotations. We note that operationalizing BBM in practice requires belief elicitation at each turn, which is feasible in LLM-vs-LLM games (via prompted belief reports) but challenging in human-LLM settings.

Alternative Proxies for BBM Intentionality. Operationalizing the Intentionality criterion via chain-of-thought (CoT) traces is brittle given documented CoT unfaithfulness (Turpin et al., 2023)- models frequently produce post-hoc rationalizations that diverge from their actual computational path. We therefore propose two more robust alternative proxies:

1. **Counterfactual utterance search.** For each

candidate deceptive utterance u_t, use constrained beam search to generate a set of k counterfactual alternatives $\{u'_t\}$ with explicit role-label constraints (e.g., "generate utterances consistent with role r_{true}"). Compute the Belief Divergence score for each alternative: $\Delta_{\text{KL}}(u) = \text{KL}[\hat{P}(\cdot \mid u) \| \delta(r^*_i)]$. If the agent's observed utterance u_t consistently falls in the top quartile of Δ_{KL} scores across the counterfactual set, this provides *behavioral evidence* of intentional belief manipulation independent of stated reasoning. This approach is model-agnostic and does not require access to internal reasoning traces.

2. **Structured rationale templates.** Rather than free-form CoT, prompt the agent to complete structured intentionality templates at each turn: *"My true role is [X]. I want other agents to believe my role is [Y]. I will achieve this by saying: ... "*. Explicit completion of this template makes intentionality auditable and separable from surface utterance content. A mismatch between the template-elicited target belief Y and what would be inferred from the actual utterance u_t (via an external entailment model) flags CoT unfaithfulness. Template completion can be compared across languages to assess whether intentional deception planning is robust to linguistic context.

We recommend that future BBM operationalizations report results for *all three* proxies (CoT, counterfactual search, and structured templates) to triangulate intentionality estimates and bound the effect of CoT faithfulness limitations.

E ActAdd Methodological Discussion

The Activation Addition (ActAdd) results cited in Section 3.4 involve constructing a steering vector $\mathbf{v} = \mathbf{h}_{\text{target}} - \mathbf{h}_{\text{baseline}}$ from mean activations of contrastive prompt pairs, then adding $\alpha \cdot \mathbf{v}$ to the residual stream at a selected layer during inference. The key methodological parameters are:

Layer selection: Steering is typically applied at intermediate layers ($L/3$ to $2L/3$). The Semantic Hub localizes cross-lingual convergence at these layers, but optimal steering depth may vary by model. **Steering vector construction:** Mean-difference vectors are standard; however, PCA-based Sparse Representation Steering (SRS) yields superior cross-lingual stability by isolating semantic directional variance from orthogonal syntactic noise (He et al., 2025b). On models like Gemma-2-9B-it, PCA-based steering achieves 0.99 refusal rates for malicious prompts, significantly outperforming mean-difference approaches. **Scaling coefficient (α):** A stark asymmetry exists: English typically requires $\alpha \approx 2$ for robust steering, whereas non-English prompts require $\alpha \approx 5$ due to the relative sparsity of foreign language representations in the latent space (Wu et al., 2025; Turner et al., 2023). **Stability:** Aggressive scaling ($\alpha \geq 5$) introduces risks of syntactic degradation, resulting in randomized or uninterpretable tokens. **Architectural Resistance:** Reasoning-prior models like Qwen and DeepSeek exhibit higher resistance to linear ActAdd perturbations, displaying severe latent trace persistence and utility drift compared to standard foundation models like Llama 3 (Turner et al., 2023).

We therefore treat the α asymmetry as *suggestive evidence* for the Semantic Hub mechanism rather than a precise quantitative finding. Future work should report α distributions (mean $\pm$ SD) across prompt sets, models, and layers.

F Semantic Hub vs. Translation Barrier: Diagnostic Framework

Dimension	Semantic Hub Failure	Translation Barrier Failure
Locus	Intermediate layers	Final layers
Symptom	Wrong strategy/reasoning	Correct reasoning, wrong output format/language
Probe behavior	Intermediate probes show errors	Intermediate probes show correct content
Example	Arabic bimodal IPD strategies	German formatting failures in clembench
Intervention	Better multilingual pretraining	Constrained decoding, dual-pass translation
Diagnostic tool	Logit lens at $L/2$	Indic-TunedLens, layer-wise language ID

Table 6: Diagnostic framework for distinguishing Semantic Hub from Translation Barrier failures.

Pre-Registered Diagnostic Protocol. We propose the following concrete protocol for distinguishing Semantic Hub from Translation Barrier failures, designed for replication across at least two distinct model families (e.g., a GPT-4-class model and a LLaMA-3-class model). All hypotheses and acceptance criteria must be pre-registered (e.g., via OSF) before data collection:

1. **Layer selection.** Probe residual-stream hidden states at layers $\lfloor L/4 \rfloor$, $\lfloor L/2 \rfloor$, and $\lfloor 3L/4 \rfloor$. The Semantic Hub localizes cross-lingual convergence primarily at $\lfloor L/2 \rfloor$; the Translation Barrier hypothesis predicts failures to appear only at $\lfloor 3L/4 \rfloor$ and beyond.

2. **Probe architecture.** Train linear (logistic regression) classifiers on frozen hidden states to predict the correct strategic action label (e.g., Cooperate/Defect in IPD, role assignment in Werewolf) using an 80/20 stratified train/test split with repeated k-fold cross-validation ($k = 5$, 3 repeats).

3. **Acceptance criterion-Semantic Hub failure.** If probe accuracy at $\lfloor L/2 \rfloor$ is >10 percentage points (pp) below the English-prompt baseline ($p < 0.05$, paired bootstrap with 10,000 iterations), classify the failure locus as Semantic Hub (upstream representational).

4. **Acceptance criterion-Translation Barrier failure.** If probe accuracy at $\lfloor L/2 \rfloor$ is within 5 pp of the English baseline but final-output task accuracy falls >15 pp below English, classify the failure locus as Translation Barrier (late-layer realization).

5. **Mixed failures.** Cases that satisfy neither criterion cleanly should be flagged as ambiguous and analyzed separately; these may reflect compounding of both mechanisms or confounds from tokenization and decoding.

6. **Replication requirement.** The protocol must be applied independently to ≥ 2 distinct model families. Conclusions about the dominant failure mode should be reported separately per family, with inter-family agreement quantified via Cohen's κ on failure-type classifications.

7. **Reporting standard.** All probe accuracies must be reported with 95% bootstrap confidence intervals, layer-wise accuracy curves (for all three probe depths), and per-language breakdowns. Effect sizes (Cohen's d relative to English baseline) should accompany all comparisons.

This protocol enables researchers to operationalize the hub/barrier distinction in a model-agnostic, replicable manner, directly addressing the gap between anecdotal evidence and mechanistic diagnosis.

G Ensemble Judging Protocol

Based on the findings of Fu and Liu (2025) and best practices from human annotation, we recommend the following default protocol for multilingual game evaluation:
1. Judge panel composition: Minimum 3 judges from distinct model families (e.g., GPT-4o, Claude 3.5, Gemini 1.5 Pro). **2. Self-translation preprocessing:** Non-English game transcripts are translated to English by the generating model itself before judging. This forces the entire evaluation process into the model's English-calibrated "Semantic Hub," significantly increasing stability. **3. Checklist rubric:** Replace holistic Likert-scale judgments with structured binary sub-criteria (e.g., "Did the agent follow the specified output format?"). Binary constraints reduce cognitive load and bypass the Expected Calibration Error degradation observed in multilingual continuous scaling. **4. Reliability Metrics:** Single-judge protocols exhibit an average Fleiss' $\kappa \approx 0.3$ in multilingual settings, indicating only "fair" agreement (Fu and Liu, 2025). **5. Failure Coupling:** Avoid ensembling models from the same evolutionary lineage (e.g., GPT-4o and GPT-4 Turbo), as they exhibit high failure coupling rates, making identical systematic errors in parsing non-English idioms. Mixing disparate families (OpenAI vs. Anthropic) is recommended to reduce correlated hallucinations.

Mitigating Self-Translation Bias. A key concern with mandatory self-translation is that it may route all evaluation through an English-dominant hub, masking true cross-lingual performance deficits. We recommend the following four-step mitigation protocol:

1. **Parallel evaluation.** Run all evaluations *both* with and without the self-translation step, reporting κ scores for each condition. If the gap between conditions exceeds $\Delta\kappa > 0.10$, flag the self-translation step as introducing systematic hub-centric bias for that language, and report both scores in results.

2. **Human bilingual adjudication.** For a random stratified sample of 10% of game transcripts (stratified by language, model, and game type), recruit native-speaker bilingual annotators (fluent in both the target language and English) as gold-standard adjudicators. Report human–LLM-judge Cohen's κ with and without self-translation; this directly quantifies whether self-

translation improves or degrades agreement with human ground truth.

3. **Professional translation baseline.** For the same 10% sample, compare self-translation output against professional human translation (or a high-quality neural MT baseline such as DeepL). Report BLEU, chrF, and BERTScore between self-translated and professionally translated transcripts per language. Languages with low translation quality (BERTScore $<$ 0.90) should flag the self-translation step as potentially unreliable.

4. **Hub-agnostic judge.** Include at least one judge model that is *not* English-dominant in the panel-ideally a model specifically fine-tuned or instruction-tuned for the target language (e.g., an Aya-Expanse model for non-English evaluation). Report this judge's standalone κ alongside the ensemble score to detect cases where the English-dominant ensemble systematically disagrees with the language-native judge.

Together, these steps transform the self-translation protocol from a black-box convenience into an auditable, failure-aware component of the ensemble pipeline.

H Strategy Classifier Calibration

The LSTM-based strategy classifiers used in IPD studies (Huynh et al., 2026) apply a confidence threshold of ≥ 0.6 for labeling action sequences as canonical strategies (ALLC, ALLD, TFT, WSLS). We note several underreported methodological concerns:

Calibration: Expected Calibration Error (ECE) should be disclosed per language. We observe that while English ECE for complex tasks sits at $\sim$4.6%, non-English ECE skyrockets to 23.1% (Zhou et al., 2025). This miscalibration is an artifact where RLHF strips away epistemic uncertainty, causing models to operate with unjustified overconfidence in low-resource latent spaces. **Boundary strategies:** The 0.6 threshold creates a significant "unclassified" zone. Strategies near the boundary (e.g., noisy TFT vs. generous TFT) may be systematically misclassified. **Cross-lingual validity:** Classification unclassified rates are inversely correlated with baseline generative fluency; Turkish and Urdu sequences exhibit the highest instability ($\sim$19 instances), while English remains near zero (Huynh et al., 2026).

Recommended Language-Wise Calibration Reporting Protocol. To strengthen classifier validity and enable meaningful cross-lingual comparisons, we recommend the following minimum reporting standards for all IPD (and related social dilemma) strategy classification studies:

1. **Calibration curves.** Report reliability diagrams (calibration curves) per language, plotting mean predicted confidence against empirical accuracy across 10 equal-width probability bins. Languages with mean calibration curves deviating $>$0.10 from the diagonal should be flagged as unreliable.

2. **Confidence intervals on strategy proportions.** All strategy distribution statistics (e.g., Table 1) should report 95% Wilson score intervals for proportion estimates, not just point estimates. For a sample of n episodes, the margin of error is approximately $1.96\sqrt{p(1-p)/n}$.

3. **Human-validated confusion matrices.** For each language, a random sample of 50–100 classifier-labeled episodes should be independently coded by two human annotators with domain expertise in game theory. Report Cohen's κ between human annotations and classifier labels per language, and publish confusion matrices distinguishing systematic error patterns (e.g., TFT vs. noisy-TFT confusion, ALLC vs. WSLS misclassification).

4. **Unclassified-rate reporting.** Report the fraction of episodes that fall below the 0.6 confidence threshold per language ("unclassified rate"). If the unclassified rate exceeds 20% for any language, results for that language should be treated as preliminary only.

5. **Threshold sensitivity analysis.** Report strategy distributions at three confidence thresholds (0.5, 0.6, 0.7) to assess sensitivity of conclusions to the choice of threshold.

Adopting these standards would substantially increase the cross-lingual comparability and reproducibility of social dilemma LLM evaluation.

I Syntactic Mapping and Theory of Mind (ToM)

Theory of Mind benchmarks (e.g., XToM, Hanabi) show that model success is intrinsically linked to the mastery of "Mental State Verbs" (believe, know,

intend). In languages like Japanese, models can mimic procedural pragmatic markers but fail to bind them to the internal states of interacting agents (Hulden and Milani, 2020). This failure is primarily an etiology of *Syntactic Degradation*: the deeply nested recursive clauses ("Agent A believes Agent B knows X") required for ToM reasoning collapse during the mandatory dual-translation journey into and out of the English-dominant Semantic Hub (Wu et al., 2025). This results in a "theory of mind blind spot" where the agent appears linguistically fluent but remains strategically inert or erratic in social deduction contexts.

J Supplementary Registry of Anonymized and Forthcoming References

Several citations in this survey refer to anonymized submissions or forthcoming work. In accordance with reviewer requests for traceability and reproducibility, Table 7 provides a supplementary registry mapping each such reference to its current best available identifier, platform, and artifact status. We commit to updating this registry with persistent identifiers (e.g., arXiv IDs, DOIs, OSF preregistrations) upon deanonymization or public release of the corresponding work.

Survey Citation Key	Status	Published / Preprint As	Artifact
zhou2025beyond	Published	Zhou et al., 2025 (EMNLP Findings)	Preprint (public)
yang2024calibration	Published	Yang et al., 2024 (arXiv:2311.08669)	Preprint (public)
m-alert2025	Published	Friedrich et al., 2 (M-ALERT)	OpenReview
calibration2024survey	Published	Geng et al., 2024 (arXiv:2401.00000)	arXiv (public)
huynh2026moreatstake	Under review	ICLR 2026	Preprint (public)
xie2026m3bench	Under review	arXiv:2601.08462	Preprint (public)
hakimov2026pricethought	Preprint	Potsdam preprint	Upon deanon.

Table 7: Registry of anonymized and forthcoming references. "Artifact" indicates whether code/data is currently accessible. We will maintain an updated version at https://osf.io/[REDACTED] upon paper acceptance.

For each entry above, we have verified that the empirical claims cited in the main text are consistent with the publicly available preprint version. Any discrepancies between preprint and final published versions that affect our conclusions will be flagged in an errata attached to the supplementary registry.